

Chapter 14: Orthogonal Vectors

Example: Let $\mathbf{u} = \begin{bmatrix} 2 \\ 0 \\ 3 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} -6 \\ 0 \\ 4 \end{bmatrix}$. What is the angle θ between $\mathbf{u}$ and $\mathbf{v}$?

$$\cos \theta = \frac{\vec{u} \cdot \vec{v}}{\|\vec{u}\| \cdot \|\vec{v}\|} = \frac{0}{\sqrt{13} \cdot \sqrt{52}} = 0$$

$$\Rightarrow \theta = \frac{\pi}{2} / 90^\circ.$$

Definition: Two vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^n$ are **orthogonal**, or **perpendicular**, if and only if $\mathbf{u} \cdot \mathbf{v} = 0$.

Example: Which pair(s) of the following three vectors are perpendicular?

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} -2 \\ -6 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} -3 \\ 1 \end{bmatrix}$$

$$\vec{v}_1 \cdot \vec{v}_2 = -2 - 18 \neq 0, \text{ not orthogonal.}$$

$$\vec{v}_1 \cdot \vec{v}_3 = -3 + 3 = 0, \text{ orthogonal}$$

$$\vec{v}_2 \cdot \vec{v}_3 = 6 + (-6) = 0, \text{ orthogonal.}$$

Example: Consider $\mathbf{v} = \begin{bmatrix} 2 \\ -3 \\ 1 \end{bmatrix}$. Find the set of all vectors which are orthogonal to $\mathbf{v}$ in $\mathbb{R}^3$.

$$2x_1 - 3x_2 + x_3 = 0.$$

Orthogonal Projection

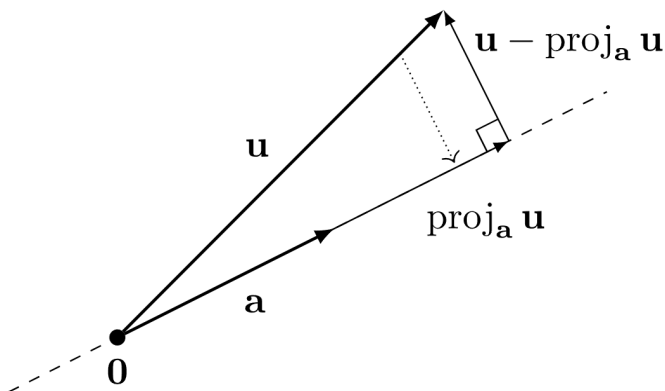
Consider two vectors $\mathbf{u} = \overrightarrow{OQ}$ and $\mathbf{a} = \overrightarrow{OR}$, which both have initial point at the origin, and terminal points at Q and R respectively. Consider the line ℓ through $\mathbf{a}$.

Question: What is the shortest distance from the terminal point Q to the line ℓ ?
put tikz here.

Definition: The **orthogonal projection** of $\mathbf{u}$ onto $\mathbf{a}$ is

$$\text{proj}_{\mathbf{a}} \mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{a}}{\|\mathbf{a}\|^2} \mathbf{a}$$

When the initial point of $\text{proj}_{\mathbf{a}} \mathbf{u}$ is placed at the origin, the terminal point will be the point closest to $\mathbf{u}$ through the origin and parallel to $\mathbf{a}$.



Fact: Vectors $\mathbf{a}$ and $\mathbf{u} - \text{proj}_{\mathbf{a}} \mathbf{u}$ are orthogonal.

Definition: The vector $\mathbf{u} - \text{proj}_{\mathbf{a}} \mathbf{u}$ is called the **vector component of $\mathbf{u}$ orthogonal to $\mathbf{a}$** .

Note: Your homework calls this the “orthogonal complement.”

Example: Find the orthogonal projection of $\begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix}$ onto $\begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}$ and shortest distance from $\begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix}$ to the line containing $\begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}$.

$$\text{Let } \vec{u} = \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix}, \quad \vec{a} = \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix},$$

$$\text{then } \text{Proj}_{\vec{a}} \vec{u} = \vec{u} \cdot \vec{a} \cdot \frac{\vec{a}}{\|\vec{a}\|^2}$$

$$= 7 \cdot \frac{\vec{a}}{14} = \frac{\vec{a}}{2} = \begin{bmatrix} 1/2 \\ -1 \\ 3/2 \end{bmatrix}$$

$$\vec{u} - \text{Proj}_{\vec{a}} \vec{u} = \begin{bmatrix} 5/2 \\ 2 \\ 1/2 \end{bmatrix}$$

$$\begin{aligned} \|\vec{u} - \text{Proj}_{\vec{a}} \vec{u}\| &= \sqrt{\frac{25}{4} + \frac{16}{4} + \frac{1}{4}} \\ &= \sqrt{\frac{21}{2}} \end{aligned}$$